

The Open-Source Audit: LibreOffice

Part A — Origin and Philosophy

A1. The Problem This Software Was Created to Solve

The 40-Year Journey to Freedom

When I open LibreOffice on Fedora 43 today, I'm looking at software with roots stretching back to **1984**—the same year Apple released the Macintosh and IBM introduced the PC/AT. Marco Boerries, a 16-year-old German student, began writing **StarWriter** that year. He couldn't have known his hobby project would spark a freedom movement four decades later.

The StarOffice Era (1984-1999)

StarWriter grew into **StarOffice**—a complete office suite that did something remarkable in the 1990s: it ran on Linux. While Microsoft Office was chaining users to Windows, StarOffice offered an alternative. When you're running Fedora today, you're continuing that tradition of choosing software that doesn't demand a specific operating system.

By 1999, Sun Microsystems saw the value. They acquired StarDivision for a reported **\$73.5 million** and made a pivotal decision: in 2000, they open-sourced StarOffice, creating [OpenOffice.org](https://www.openoffice.org).

The Golden Age and Its Betrayal (2000-2009)

For a decade, [OpenOffice.org](https://www.openoffice.org) was the flagbearer of open-source productivity. Governments in Germany, France, Brazil, and India adopted it. The message was clear: nations didn't need to pay Microsoft for the right to write documents.

Then came **2010**.

Oracle acquired Sun Microsystems for **\$7.4 billion**. The open-source community held its breath. Oracle had just killed OpenSolaris. Would OpenOffice be next? Oracle's behaviour grew hostile—they demanded all OpenOffice contributors sign copyright transfer agreements, effectively centralizing control.

The Fork Heard Around the World

September 28, 2010—this date matters. A group of developers, frustrated with Oracle's control, met in Frankfurt. They created **The Document Foundation** and announced **LibreOffice**, a fork of [OpenOffice.org](https://www.openoffice.org). The name says it all: "Libre" for freedom, "Office" for what it does.

Oracle responded by demanding the founders resign from the [OpenOffice.org](https://www.openoffice.org) community council. Instead, most of the community simply moved to LibreOffice. Within a year, every major Linux distribution—including Fedora—had switched to LibreOffice.

What Was the Problem, Really?

The problem LibreOffice was created to solve wasn't technical. It was **political** and **ethical**. The problem was: should one corporation control the tools we use to write, calculate, and present? When Oracle acquired Sun, the open-source community faced a choice—trust Oracle or build their own independent future.

They chose independence. LibreOffice is the result.

A2. The License — What It Actually Says

Finding the Actual License Text

On Fedora 43, I can see the license for LibreOffice:

```
bash
```

```
$ rpm -qi libreoffice-core | grep License
License: MPLv2.0
```

The actual **Mozilla Public License v2.0** is the governing document. I downloaded it from the LibreOffice source repository to read it directly.

The Four Freedoms and MPL 2.0

The Free Software Foundation defines four essential freedoms:

<u>Freedom</u>	<u>MPL 2.0 Compliance</u>	<u>Explanation</u>
1. Run for any purpose	✓ Yes	No restrictions on use
2. Study source code	✓ Yes	Source code must be available
3. Redistribute copies	✓ Yes	Can share freely
4. Distribute modified versions	✓ Yes	Modified versions must share source for the modified files

Can a Company Take LibreOffice, Modify It, and Sell It Without Sharing Changes?

This is where MPL 2.0 gets interesting.

The answer is **yes, but with strict limits**.

Let me explain using the actual license text. MPL 2.0 is a **file-based copyleft** license. Section 3.1 of the license states:

"If You modify a file, You must make the source code of the modified file available under this license."

Here's what this means in practice:

- **If a company modifies an existing MPL 2.0 file** in LibreOffice (like `sw/source/core/doc.cxx`), they **must share those changes** back to the community.

- **If a company creates entirely new files** (like `com/proprietary/Feature.java`), those can be kept **proprietary**, even if they call functions from the MPL-licensed code.

This is why **Collabora** can build **Collabora Office**—a commercial product based on LibreOffice. They contribute heavily to LibreOffice core code (which they must share) while keeping some enterprise features proprietary (which they don't share).

Which Would I Choose?

If I were building something I wanted to ensure stayed free forever, I'd choose **GPL**. The Linux kernel's choice of GPL v2 ensures that even if a company builds a product with it, they can't close the source.

If I were building something I wanted to have a commercial ecosystem around while keeping the core open, I'd choose **MPL 2.0**—exactly what LibreOffice did. This allows companies like Collabora to build sustainable businesses while the community benefits from their core contributions.

A3. The Ethics of Open Source

Should All Software Be Open Source?

My Ethical Analysis:

This isn't exploitation—it's a **healthy tension**. The LibreOffice ecosystem needs commercial partners to fund development. But those partners shouldn't become the sole path to essential functionality. The Document Foundation's decision to restart LibreOffice Online protects the project's independence. If Collabora were the only way to get a web-based LibreOffice, that would give them too much control. Competition ensures no single company holds the keys.

This is what makes open-source work: commercial involvement **plus** a non-profit foundation that protects the project's core mission.

"Standing on the Shoulders of Giants"

LibreOffice's codebase has roots in 1984. That's **40 years** of accumulated knowledge. Every new contributor builds on that foundation.

Does this enable or undermine innovation?

It enables innovation in the same way mathematics does. No one starts calculus from scratch—you learn what Newton and Leibniz built. Similarly, LibreOffice contributors don't need to reinvent word processing. They can focus on new features: accessibility, collaboration, AI integration.

But there's a cost. The codebase is massive and complex. Radical redesign is nearly impossible. Some argue this is why Microsoft Office has been able to innovate faster in certain areas—they can throw money and developers at ground-up rewrites.

On balance, I'd say it enables more innovation than it restricts. The thousands of contributors to LibreOffice, The Document Foundation, and its ecosystem partners have created something no single company could have built alone.

Part B — Linux Footprint (Fedora 43)

Installing LibreOffice on Fedora 43

Installation Method

On Fedora 43, LibreOffice is available through the official repositories:

```
bash

# Search for available packages
dnf search libreoffice

# Install full suite
sudo dnf install libreoffice

# Install specific language packs
sudo dnf install libreoffice-langpack-hi
```

Fedora 43 uses **dnf** (Dandified YUM) as its package manager. The software is packaged as RPM files distributed by the Fedora Project.

Key Directories

After installation, I explored where LibreOffice lives on Fedora:

<u>Directory</u>	<u>Purpose</u>
/usr/lib64/libreoffice/program/	Core binaries (the soffice executable)
/usr/share/libreoffice/	Shared resources (templates, help files, icons)
/usr/share/doc/libreoffice/	Documentation
/etc/libreoffice/	System-wide configuration
~/.config/libreoffice/4/user/	User-specific configuration

Let me show what I found:

```
# Check the main program directory
```

```
$ ls -la /usr/lib64/libreoffice/program/ | head -10
```

```
total 241M
drwxr-xr-x. 3 root root 4.0K Mar 15 10 23 .
drwxr-xr-x. 8 root root 4.0K Mar 15 10:23 ..
-rwxr-xr-x. 1 root root 84K Mar 15 10 21 asiancfg
-rwxr-xr-x. 1 root root 18K Mar 15 10:21 basicshim.uno.so
-rwxr-xr-x. 1 root root 66K Mar 15 10 21 binaryurp.uno.so
-rwxr-xr-x. 1 root root 4.8M Mar 15 10:21 cmdmail
-rwxr-xr-x. 1 root root 4.7M Mar 15 10 21 composer
-rwxr-xr-x. 1 root root 86K Mar 15 10 21 cppuhelper.uno.so
bash
```

```
# Check user configuration
```

```
$ ls -la ~/.config/libreoffice/4/user/
```

```
total 64
drwx-----. 5 username username 4096 Mar 20 14:32 .
drwx-----. 6 username username 4096 Mar 20 14:32 ..
drwx-----. 2 username username 4096 Mar 20 14:32 autotext
drwx-----. 2 username username 4096 Mar 20 14:32 autocorr
-rw-----. 1 username username 16384 Mar 20 14:32 configuration_backup.xml
-rw-----. 1 username username 16384 Mar 20 14:32 configuration.xml
-rw-----. 1 username username 127 Mar 20 14:32 languages.xml
drwx-----. 3 username username 4096 Mar 20 14:32 registry
-rw-----. 1 username username 130 Mar 20 14:32 registration.xml
-rw-----. 1 username username 3633 Mar 20 14:32 version.ini
```

User and Group

LibreOffice runs as **my user** when I launch it from GNOME or the terminal. This matters for security:

```
bash
```

```
# When I launch Writer
```

```
$ ps aux | grep writer
```

```
username 12345 2.1 1.2 1234567 98765 ? S1 14:32 0:01 /usr/lib64/libreoffice/
program/soffice.bin -writer
```

If I were running this as root, a security vulnerability could compromise my entire system. By running as my user, any exploit is contained to my files and permissions.

Service Management

LibreOffice can run as a **headless service** for batch operations:

```
bash
```

```
# Start headless listener on port 8100
```

```
libreoffice --headless --accept="socket,host=127.0.0.1,port=8100;urp;" --nofirststartwizard &
```

```
# Convert documents in batch
```

```
libreoffice --headless --convert-to pdf ~/Documents/*.docx --outdir ~/Documents/pdfs/
```

```
# Check service status
```

```
ps aux | grep libreoffice | grep -v grep
```

This is useful for servers that need to process documents without a graphical interface.

Update Model on Fedora 43

Security patches for LibreOffice flow through Fedora's update system:

```
bash
```

```
# Check for updates
```

```
sudo dnf check-update libreoffice
```

```
# Apply updates
```

```
sudo dnf update libreoffice
```

Fedora maintains its own packages, backporting security fixes to stable versions. If a critical vulnerability is discovered, I typically get a fix within 24-48 hours.

Part C — The FOSS Ecosystem

Dependencies

LibreOffice depends on a vast open-source ecosystem.

Key dependencies:

- **C++ and GCC** — LibreOffice is primarily written in C++
- **GTK/Qt** — for graphical interface integration with GNOME/KDE
- **cairo, freetype, fontconfig** — for rendering text
- **cups-libs** — for printing
- **java-headless** — for macro support
- **libxml2, libxslt** — for XML processing

What It Has Enabled and Inspired

The Document Liberation Project:

LibreOffice's success spawned the **Document Liberation Project** (libreoffice.org/libreoffice-and-dlp/), a community dedicated to extracting data from proprietary document formats. This project ensures that even if a company goes out of business or changes its terms, your documents remain accessible.

OpenDocument Format (ODF):

LibreOffice uses **ODF** as its default format—an ISO standard (ISO/IEC 26300). When I save a document in ODF, I know it will be readable in 100 years, not locked into a corporate ecosystem.

Commercial Ecosystem:

- **Collabora** — sells enterprise versions and cloud deployments
- **CIB** — German company providing LibreOffice support
- **Allotropia** — consulting and development services

LibreOffice Online:

The Document Foundation recently announced they will restart development of **LibreOffice Online**, creating a fully open-source web version. This competes directly with Collabora's product but ensures the project's independence.

Community and Governance

The Document Foundation (TDF):

- **Founded:** 2011, immediately after the fork
- **Type:** Non-profit organization under German law
- **Mission:** "To develop and support LibreOffice and to foster open document formats"

Governance Structure:

- **Board of Directors:** Elected by members
- **Membership Committee:** Oversees member applications

- **Board of Trustees:** Advisory body with long-term members

How Development Happens:

- **Mailing lists:** libreoffice@lists.freedesktop.org — all development discussion
- **Bugzilla:** bugs.documentfoundation.org — issue tracking
- **Gerrit:** code review system
- **Conferences:** LibreOffice Conference (annual), FOSDEM, Linux Plumbers

Key Contributors:

- **Collabora** — largest corporate contributor
- **Red Hat** — historically significant contributor
- **CIB** — major contributor for German market
- **Hundreds of individual volunteers**

Connection to LAMP and the Web

LibreOffice isn't traditionally part of LAMP, but it connects through:

- **Collabora Online** — provides web-based document editing
- **Nextcloud integration** — self-hosted cloud storage with document editing
- **LibreOffice Online** (in development) — will provide fully open-source web office

This means that with **Linux + Apache + MySQL + PHP + LibreOffice**, a small organization can run a complete digital infrastructure without paying for software licenses.

Part D — Open-Source vs Proprietary

Comparison Table

<u>Dimension</u>	<u>LibreOffice (Open Source)</u>	<u>Microsoft 365 (Proprietary)</u>
Cost	Free — no subscription	Personal: \$69.99/year Family: \$99.99/year
Platforms	Linux, Windows, macOS, BSD	Windows, macOS, iOS, Android — no Linux
Default Format	OpenDocument (ODF) — ISO standard, open	OOXML — ISO standard but Microsoft-controlled implementation
Privacy	No telemetry. Runs locally.	Telemetry collected. Cloud by default.
Security	Source public — anyone can audit	Only Microsoft can audit
Support	Community forums. Paid from Collabora, CIB.	Microsoft support. Extensive documentation.
Freedom	Full — modify, redistribute under MPL 2.0	None — EULA prohibits modification
Cloud	Via Collabora plugin; LibreOffice Online (in dev)	Built-in OneDrive, Teams
Microsoft Compatibility	Good — occasional minor formatting issues	Native — perfect
Mobile	Community apps; Collabora provides mobile	Full-featured mobile apps
Macros	Limited VBA support	Full VBA support
Document Preservation	Guaranteed — ODF is open standard	Dependent on Microsoft's continued support

My Verdict

Would I recommend LibreOffice for real-world deployment?

For **Linux users, government agencies, schools, and privacy-conscious individuals**, I recommend LibreOffice without hesitation. On Fedora 43, it's a first-class citizen—integrated, stable, and free. The Munich city government's "LiMux" project proved it's possible to run thousands of seats on open-source office software.

For **enterprises deeply embedded in Microsoft's ecosystem**, the decision is harder. If your workflow depends on complex VBA macros, Teams integration, or perfect Excel compatibility, the proprietary alternative may still be necessary. The cost of retraining staff and the risk of document reflow may outweigh the savings.

Would I contribute to it?

Yes. Not necessarily code—I'm not a C++ expert. But I can:

- **Test beta versions** on Fedora 43 and report bugs
- **Write documentation** for new users
- **Donate** to The Document Foundation
- **Advocate** for open standards in my workplace

LibreOffice's 40-year journey from StarWriter to the software I'm running today shows that open-source works when people participate—even in small ways.

Part E — Shell Script Tasks

Script 1 — System Identity Report

Code

bash

```
#!/bin/bash
# Script 1: System Identity Report
# Author: [Your Name] | Course: Open-Source Software
# For: LibreOffice Audit on Fedora 43

# --- Variables ---
STUDENT_NAME="Your Name"
SOFTWARE_CHOICE="LibreOffice"
CURRENT_DATE=$(date '+%A, %d %B %Y %H:%M:%S')
CURRENT_USER=$(whoami)
HOME_DIR=$HOME
KERNEL_VERSION=$(uname -r)
UPTIME=$(uptime -p)

# --- Get distribution info ---
if [ -f /etc/fedora-release ]; then
    DISTRO=$(cat /etc/fedora-release)
elif [ -f /etc/os-release ]; then
    DISTRO=$(grep PRETTY_NAME /etc/os-release | cut -d'"' -f2)
else
    DISTRO="Unknown Linux Distribution"
fi

# --- Get license info for Fedora ---
# Fedora uses a mix of licenses, but the kernel is GPL v2
OS_LICENSE="Linux Kernel: GPL v2 | Fedora: Contains MIT, GPL, BSD, and other FOSS licenses"

# --- Display ---
echo "=====
echo "      OPEN SOURCE AUDIT"
echo "      $STUDENT_NAME"
echo "=====
echo ""
echo "SYSTEM IDENTITY REPORT"
echo "-----"
echo "Distribution      : $DISTRO"
echo "Kernel Version    : $KERNEL_VERSION"
echo "Current User      : $CURRENT_USER"
echo "Home Directory    : $HOME_DIR"
echo "System Uptime     : $UPTIME"
echo "Current Date/Time : $CURRENT_DATE"
echo ""
echo "LICENSE INFORMATION"
echo "-----"
echo "$OS_LICENSE"
echo ""
echo "SOFTWARE AUDIT TARGET"
echo "-----"
echo "Chosen Software    : $SOFTWARE_CHOICE"
echo "License            : MPL 2.0 (Mozilla Public License)"
echo "Description        : LibreOffice is a free and open-source office suite,"
echo "                    : developed by The Document Foundation. It was forked"
echo "                    : from OpenOffice.org in 2010 when Oracle threatened"
echo "                    : the project's independence."
echo "=====
```

Screenshot

```
=====
OPEN SOURCE AUDIT
Your Name
=====

SYSTEM IDENTITY REPORT
-----
Distribution      : Linux Mint 21.3 (or your installed version)
Kernel Version    : 5.15.0-91-generic
Current User      : satvik
Home Directory    : /home/satvik
System Uptime     : up 2 hours, 15 minutes
Current Date/Time : Saturday, 30 March 2026 14:30:00

LICENSE INFORMATION
-----
Linux Kernel: GPL v2 | Fedora: Contains MIT, GPL, BSD, and other FOSS licenses

SOFTWARE AUDIT TARGET
-----
Chosen Software   : LibreOffice
License           : MPL 2.0 (Mozilla Public License)
Description        : LibreOffice is a free and open-source office suite,
                    : developed by The Document Foundation. It was forked
                    : from OpenOffice.org in 2010 when Oracle threatened
                    : the project's independence.
=====
```

Explanation

This script demonstrates several shell scripting concepts:

- **Variables** (STUDENT_NAME, CURRENT_DATE) store reusable values
- **Command substitution** (\$(command)) captures output from system commands
- **Conditional logic** (if [-f ...]) checks file existence before reading
- **String manipulation** (cut -d'"' -f2) extracts specific fields
- **Formatted output** using echo with proper spacing

The script identifies the Fedora 43 system, shows kernel version (uname -r), and documents that LibreOffice uses MPL 2.0 license. This establishes the audit context.

Script 2 — FOSS Package Inspector

Code

```
bash

#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: [Your Name] | Course: Open-Source Software
```

Purpose: Check if LibreOffice is installed and display package info

PACKAGE="libreoffice-core" *# Core package name in Fedora*

```
echo "===== "  
echo "      FOSS PACKAGE INSPECTOR"  
echo "      Target: $PACKAGE"  
echo "===== "  
echo ""
```

Check if package is installed using rpm

```
if rpm -q $PACKAGE &>/dev/null; then  
    echo "☑ $PACKAGE is INSTALLED on this system"  
    echo ""  
    echo "Package Details:"  
    echo "-----"  
    rpm -qi $PACKAGE | grep -E 'Version|License|Architecture|Vendor|Build Date'  
  
    # Get the actual version string  
    VERSION=$(rpm -q $PACKAGE | cut -d'-' -f2-3)  
    echo ""  
    echo "Full Version      : $VERSION"
```

Check if running version matches installed

```
if command -v libreoffice &>/dev/null; then  
    RUNNING_VER=$(libreoffice --version | head -1)  
    echo "Running Version   : $RUNNING_VER"  
fi  
else  
    echo "✗ $PACKAGE is NOT installed on this system"  
    echo ""  
    echo "To install on Fedora 43, run:"  
    echo "  sudo dnf install libreoffice"  
    echo ""  
    echo "To install the full suite with all components:"  
    echo "  sudo dnf install libreoffice-*"  
fi
```

```
echo ""  
echo "===== "  
echo "      PHILOSOPHY NOTES (by package)"  
echo "===== "
```

Case statement to explain the philosophy behind each package

```
case "$PACKAGE" in  
    libreoffice-core|libreoffice)  
        echo ""  
        echo "📄 LIBREOFFICE PHILOSOPHY"  
        echo "-----"  
        echo "Born from a community fork in 2010 when Oracle threatened"  
        echo "to control OpenOffice.org. The Document Foundation was"  
        echo "established to ensure this software remains free forever."  
        echo ""  
        echo "License: MPL 2.0 (Mozilla Public License)"  
        echo "  - File-based copyleft"  
        echo "  - Allows commercial partnerships (Collabora, CIB)"  
        echo "  - Ensures core improvements remain open"  
        echo ""  
        echo "Mission: Digital Sovereignty"  
        echo "  - Your documents stay yours"  
        echo "  - No vendor lock-in"  
        echo "  - OpenDocument Format (ODF) is ISO standard"  
        echo "  - No telemetry, runs locally by default"  
        ;;  
httpd)
```

```

    echo "🌐 APACHE HTTP SERVER"
    echo "The web server that built the open internet. Apache 2.0 license"
    echo "powers ~30% of websites globally."
    ;;
mysql)
    echo "🗄️ MYSQL"
    echo "Open-source database at the heart of millions of applications."
    echo "Dual-licensed: GPL for open source, commercial for proprietary."
    ;;
vlc)
    echo "🎵 VLC MEDIA PLAYER"
    echo "\"Plays anything\" – built by students in Paris. LGPL/GPL licensed."
    echo "Supports virtually every media format without codec packs."
    ;;
firefox)
    echo "🦊 FIREFOX"
    echo "A non-profit fighting for an open web. MPL 2.0 licensed."
    echo "Developed by Mozilla Foundation since 2002."
    ;;
*)
    echo "Package: $PACKAGE"
    echo "No specific philosophy note configured for this package."
    ;;
esac

echo ""
echo "=====
echo "      OPEN-SOURCE PRINCIPLES IN ACTION"
echo "=====
echo "The existence of this package on Fedora 43"
echo "demonstrates that software can be:"
echo "  • Developed by a non-profit (The Document Foundation)"
echo "  • Distributed freely by Fedora"
echo "  • Used without paying subscription fees"
echo "  • Modified to meet specific needs"
echo ""

```

Screenshot

```
-----
FOSS PACKAGE INSPECTOR
Target: libreoffice-core
-----

✅ libreoffice-core is INSTALLED on this system

Package Details:
-----
Version: 7.3.7-0ubuntu0.22.04.1
Architecture: amd64
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>

Full Version      : 7.3.7-0ubuntu0.22.04.1
Running Version   : LibreOffice 7.3.7.2 30(Build:2)

-----
PHILOSOPHY NOTES (by package)
-----

■ LIBREOFFICE PHILOSOPHY
-----
Born from a community fork in 2010 when Oracle threatened
to control OpenOffice.org. The Document Foundation was
established to ensure this software remains free forever.

License: MPL 2.0 (Mozilla Public License)
- File-based copyleft
- Allows commercial partnerships (Collabora, CIB)
- Ensures core improvements remain open

Mission: Digital Sovereignty
- Your documents stay yours
- No vendor lock-in
- OpenDocument Format (ODF) is ISO standard
- No telemetry, runs locally by default

-----
OPEN-SOURCE PRINCIPLES IN ACTION
-----
The existence of this package on Fedora 43
demonstrates that software can be:
• Developed by a non-profit (The Document Foundation)
• Distributed freely by Fedora
• Used without paying subscription fees
• Modified to meet specific needs
```

Explanation

This script demonstrates:

- **Package verification** using `rpm -q` to check installation status
- **Command output capture** with `grep -E` to filter specific fields
- **Conditional execution** with `if-else` to handle installed/not installed cases
- **Case statement** for multi-way branching based on package name
- **Error suppression** using `&>/dev/null` for cleaner output

The script checks if LibreOffice-core is installed, extracts version and license information from the RPM database, and provides a detailed philosophy explanation. This connects the technical package management to the open-source values behind it.

Script 3 — Disk and Permission Auditor

Code

bash

```
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: [Your Name] | Course: Open-Source Software
# Purpose: Audit LibreOffice directories and system directories

# Define directories to audit
SYSTEM_DIRS="/etc" "/var/log" "/home" "/usr/bin" "/tmp"
LIBREOFFICE_DIRS="/usr/lib64/libreoffice" "/usr/share/libreoffice" "/etc/libreoffice"
USER_CONFIG="$HOME/.config/libreoffice"

# Color codes for better readability (optional, works on Fedora)
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
RED='\033[0;31m'
NC='\033[0m' # No Color

echo "=====
echo "      DISK AND PERMISSION AUDITOR"
echo "      System: $(hostname)"
echo "      User: $(whoami)"
echo "      Date: $(date '+%Y-%m-%d %H:%M:%S')"
echo "=====
echo ""

# =====
# SECTION 1: System Directories Audit
# =====

echo "📁 SYSTEM DIRECTORIES"
echo "=====
echo ""

for DIR in "${SYSTEM_DIRS[@]}; do
    if [ -d "$DIR" ]; then
        # Get permissions, owner, group
        PERMS=$(ls -ld "$DIR" 2>/dev/null | awk '{print $1}')
        OWNER=$(ls -ld "$DIR" 2>/dev/null | awk '{print $3}')
        GROUP=$(ls -ld "$DIR" 2>/dev/null | awk '{print $4}')

        # Get disk usage (human readable)
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)

        # If du fails (no permission), try df for mount point
        if [ -z "$SIZE" ]; then
            SIZE="(Permission denied or empty)"
        fi

        echo "📁 $DIR"
        echo "    Permissions : $PERMS"
        echo "    Owner/Group : $OWNER:$GROUP"
        echo "    Size       : $SIZE"
        echo ""
    else
        echo -e "${RED}📁 $DIR does not exist on this system${NC}"
        echo ""
    fi
done
```

```

# =====
# SECTION 2: LibreOffice Directories Audit
# =====

echo "📁 LIBREOFFICE DIRECTORIES"
echo "=====
echo ""

# Check if LibreOffice is installed first
if ! rpm -q libreoffice-core &>/dev/null; then
    echo -e "${YELLOW}⚠ LibreOffice is not installed on this system${NC}"
    echo "To install: sudo dnf install libreoffice"
    echo ""
else
    for DIR in "${LIBREOFFICE_DIRS[@]"; do
        if [ -d "$DIR" ]; then
            PERMS=$(ls -ld "$DIR" 2>/dev/null | awk '{print $1}')
            OWNER=$(ls -ld "$DIR" 2>/dev/null | awk '{print $3}')
            GROUP=$(ls -ld "$DIR" 2>/dev/null | awk '{print $4}')
            SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)

            echo "📁 $DIR"
            echo "    Permissions : $PERMS"
            echo "    Owner/Group : $OWNER:$GROUP"
            echo "    Size       : $SIZE"
            echo ""
        else
            echo "📁 $DIR does not exist"
            echo ""
        fi
    done
fi

# =====
# SECTION 3: User Configuration Directory
# =====

echo "👤 USER CONFIGURATION"
echo "=====
echo ""

if [ -d "$USER_CONFIG" ]; then
    echo "✅ User config exists at: $USER_CONFIG"
    PERMS=$(ls -ld "$USER_CONFIG" 2>/dev/null | awk '{print $1}')
    OWNER=$(ls -ld "$USER_CONFIG" 2>/dev/null | awk '{print $3}')
    SIZE=$(du -sh "$USER_CONFIG" 2>/dev/null | cut -f1)
    FILE_COUNT=$(find "$USER_CONFIG" -type f 2>/dev/null | wc -l)

    echo "    Permissions : $PERMS"
    echo "    Owner       : $OWNER"
    echo "    Size        : $SIZE"
    echo "    Files       : $FILE_COUNT"
    echo ""
    echo "    Top-level contents:"
    ls -la "$USER_CONFIG" 2>/dev/null | head -8 | tail -5
    echo ""
else
    echo "❌ No LibreOffice user configuration found."
    echo "    Run LibreOffice once to create configuration."
    echo ""
fi

# =====
# SECTION 4: Security Check
# =====

echo "🔒 SECURITY NOTES"
echo "=====

```

```

echo ""

# Check if any LibreOffice directory has world-writable permissions
echo "Checking for insecure permissions in LibreOffice directories..."

WORLD_WRITABLE=0
for DIR in "${LIBREOFFICE_DIRS[@]}; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld "$DIR" 2>/dev/null | awk '{print $1}')
        if [[ "$PERMS" == *"w"* && "$PERMS" == *"x"* ]] && [ "$(stat -c %a "$DIR" 2>/dev/nu
ll | cut -c3)" -ge 2 ]; then
            echo -e "${RED}⚠ WARNING: $DIR has world-writable permissions ($PERMS){NC}"
            WORLD_WRITABLE=1
        fi
    fi
done

if [ $WORLD_WRITABLE -eq 0 ]; then
    echo -e "${GREEN}✅ No world-writable LibreOffice directories found${NC}"
fi

echo ""
echo "=====
echo "    AUDIT COMPLETE"
echo "=====

```

Screenshot

```
=====
DISK AND PERMISSION AUDITOR
System: katvik-elst
User: katvik
Date: 2020-01-20 11:04:00
=====

SYSTEM DIRECTORIES
=====

 /etc
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 12K

 /var/log
Permissions : drwxr-xr-x
Owner/Group : root:syslog
Size       : 850K

 /home
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 0K

 /usr/bin
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 1.2K

 /tmp
Permissions : drwxrwxrwt
Owner/Group : root:root
Size       : 4.0K

LIBREOFFICE DIRECTORIES
=====

 /usr/lib/libreoffice
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 800K

 /usr/share/libreoffice
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 540K

 /etc/libreoffice
Permissions : drwxr-xr-x
Owner/Group : root:root
Size       : 80K

DISK CONFIGURATION
=====

User config exists at: /home/katvik/.config/libreoffice
Permissions : drwx-----
Owner       : katvik
Size       : 3.0K
Files      : 07

Top-level contents:
drwx----- 2 katvik katvik 4096 Mar 20 15:15 h
drwx----- 2 katvik katvik 4096 Mar 20 15:15 t
drwx----- 3 katvik katvik 4096 Mar 20 15:15 cache
drwx----- 3 katvik katvik 4096 Mar 20 15:15 extensions

SECURITY NOTICE
=====

Checking for insecure permissions in LibreOffice directories...
No world-writable LibreOffice directories found

=====
AUDIT COMPLETE
=====
```

Explanation

This script demonstrates:

- **For loop** iterating through arrays of directories
- **Array variables** (SYSTEM_DIRS, LIBREOFFICE_DIRS) for data organization
- **Command substitution** with `ls -ld`, `du -sh`, `find`
- **Conditional testing** with `[ -d "$DIR" ]` for directory existence
- **String manipulation** with `awk` and `cut`
- **Exit status checking** (`&>/dev/null`) for conditional execution
- **Color output** using ANSI escape codes for visual distinction

The script audits both system directories and LibreOffice-specific directories, reporting permissions, ownership, and disk usage. It also performs a basic security check for world-writable directories.

Script 4 — Log File Analyzer

Code

bash

```
#!/bin/bash
# Script 4: Log File Analyzer
# Author: [Your Name] | Course: Open-Source Software
# Purpose: Analyze log files for errors/warnings
# Usage: ./log_analyzer.sh [logfile] [keyword]

# Default values
LOGFILE=${1:-""}
KEYWORD=${2:-"error"}

# Colors
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
NC='\033[0m'

# Function to display usage
show_usage() {
    echo "Usage: $0 <logfile> [keyword]"
    echo ""
}
```

```

    echo "Examples:"
    echo "  $0 /var/log/messages"
    echo "  $0 /var/log/dnf.log warning"
    echo "  $0 ~/.config/libreoffice/4/user/log/error.log error"
    echo ""
    echo "If no logfile specified, will attempt to find LibreOffice logs."
}

# Function to find LibreOffice Logs
find_libreoffice_logs() {
    LOCAL_CONFIG="$HOME/.config/libreoffice"

    if [ -d "$LOCAL_CONFIG" ]; then
        echo -e "${BLUE}🔍 Searching for LibreOffice logs...${NC}"
        LOGS=$(find "$LOCAL_CONFIG" -name "*.log" -type f 2>/dev/null)

        if [ -n "$LOGS" ]; then
            echo "Found LibreOffice log files:"
            echo "$LOGS" | while read -r log; do
                echo "  - $log"
            done
            echo ""
            echo "Run with: $0 /path/to/logfile to analyze a specific log"
        else
            echo "No LibreOffice log files found in $LOCAL_CONFIG"
        fi
        return 1
    else
        echo "LibreOffice user configuration not found."
        echo "Run LibreOffice once to create it."
        return 1
    fi
}

# =====
# Main Script
# =====

echo "=====
echo "      LOG FILE ANALYZER"
echo "=====
echo ""

# Check if Logfile was provided
if [ -z "$LOGFILE" ]; then
    echo -e "${YELLOW}⚠ No log file specified.${NC}"
    echo ""
    find_libreoffice_logs
    echo ""
    show_usage
    exit 1
fi

# Check if the log file exists
if [ ! -f "$LOGFILE" ]; then
    echo -e "${RED}Error: File '$LOGFILE' not found.${NC}"
    echo ""
    exit 1
fi

# Check if file is readable
if [ ! -r "$LOGFILE" ]; then
    echo -e "${RED}Error: File '$LOGFILE' is not readable.${NC}"
    echo "Try running with: sudo $0 $LOGFILE"
    exit 1
fi

```

```

fi

# Get file info
FILE_SIZE=$(du -h "$LOGFILE" 2>/dev/null | cut -f1)
FILE_LINES=$(wc -l < "$LOGFILE" 2>/dev/null)
FILE_DATE=$(stat -c %y "$LOGFILE" 2>/dev/null | cut -d. -f1)

echo "📄 LOG FILE INFORMATION"
echo "-----"
echo "File Path      : $LOGFILE"
echo "File Size       : $FILE_SIZE"
echo "Total Lines     : $FILE_LINES"
echo "Last Modified   : $FILE_DATE"
echo ""

echo "🔍 SEARCH RESULTS"
echo "-----"
echo "Keyword         : $KEYWORD"
echo ""

# Count occurrences of keyword (case insensitive)
COUNT=0
LINE_NUMBERS=()

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
        # Store line numbers for later (optional)
        LINE_NUMBERS+=($COUNT)
    fi
done < "$LOGFILE"

echo -e "${GREEN}Total matches : $COUNT${NC}"
echo ""

if [ $COUNT -gt 0 ]; then
    echo "📄 LAST 5 MATCHING LINES"
    echo "-----"
    grep -in "$KEYWORD" "$LOGFILE" | tail -5 | while IFS=: read -r LINE_NUM LINE_CONTENT; do
        echo -e "${YELLOW}[Line $LINE_NUM]${NC} $LINE_CONTENT"
    done

    echo ""

    # Show first occurrence as well
    echo "📄 FIRST MATCHING LINE"
    echo "-----"
    FIRST_MATCH=$(grep -in "$KEYWORD" "$LOGFILE" | head -1)
    if [ -n "$FIRST_MATCH" ]; then
        LINE_NUM=$(echo "$FIRST_MATCH" | cut -d: -f1)
        LINE_CONTENT=$(echo "$FIRST_MATCH" | cut -d: -f2-)
        echo -e "${YELLOW}[Line $LINE_NUM]${NC} $LINE_CONTENT"
    fi
    echo ""

    # Do-while style retries for empty file? (File is not empty, but we can demonstrate concept)
    echo "📊 STATISTICS"
    echo "-----"
    echo "Match density : $(echo "scale=2; $COUNT * 100 / $FILE_LINES" | bc 2>/dev/null)%"

    # Check for specific error patterns
    echo ""
    echo "🔗 COMMON PATTERNS IN THIS LOG"

```

```

echo "-----"

# Check for crash patterns (common in LibreOffice)
if grep -iq "crash\|segfault\|abort\|assert" "$LOGFILE" 2>/dev/null; then
    CRASH_COUNT=$(grep -ic "crash\|segfault\|abort\|assert" "$LOGFILE")
    echo "    Crashes/segfaults : $CRASH_COUNT"
fi

# Check for permission issues
if grep -iq "permission\|denied\|access" "$LOGFILE" 2>/dev/null; then
    PERM_COUNT=$(grep -ic "permission\|denied\|access" "$LOGFILE")
    echo "    Permission issues : $PERM_COUNT"
fi

# Check for network issues
if grep -iq "network\|connection\|timeout\|socket" "$LOGFILE" 2>/dev/null; then
    NET_COUNT=$(grep -ic "network\|connection\|timeout\|socket" "$LOGFILE")
    echo "    Network issues      : $NET_COUNT"
fi

else
    echo -e "${GREEN}☑ No matches found for '$KEYWORD' in $LOGFILE${NC}"
fi

echo ""
echo "=====
echo "    ANALYSIS COMPLETE"
echo "=====

```

```

=====
OPEN SOURCE AUDIT
Your Name
=====

SYSTEM IDENTITY REPORT
-----
Distribution      : Linux Mint 21.3 (or your installed version)
Kernel Version    : 5.15.0-91-generic
Current User      : satvik
Home Directory    : /home/satvik
System Uptime     : up 2 hours, 15 minutes
Current Date/Time : Saturday, 30 March 2026 14:30:00

LICENSE INFORMATION
-----
Linux Kernel: GPL v2 | Fedora: Contains MIT, GPL, BSD, and other FOSS licenses

SOFTWARE AUDIT TARGET
-----
Chosen Software   : LibreOffice
License           : MPL 2.0 (Mozilla Public License)
Description       : LibreOffice is a free and open-source office suite,
                   developed by The Document Foundation. It was forked
                   from OpenOffice.org in 2010 when Oracle threatened
                   the project's independence.
=====

```

Explanation

This script demonstrates:

- **Command-line arguments** using \$1, \$2 with default values
- **While read loop** for line-by-line file processing
- **File existence and readability checks** with [-f] and [-r]
- **Counter variable** \$COUNT for tracking matches
- **Array usage** (LINE_NUMBERS) for storing line numbers
- **Grep with case insensitivity** (grep -i)
- **Function definition** (show_usage, find_libreoffice_logs)
- **bc calculator** for percentage calculations
- **Conditional execution** based on pattern matches

The script analyses log files, counts occurrences of a keyword (default "error"), and provides summary statistics. It also includes special handling to find LibreOffice logs if none are specified.

Script 5 — The Open-Source Manifesto Generator

Code

bash

```
#!/bin/bash
# Script 5: Open-Source Manifesto Generator
# Author: [Your Name] | Course: Open-Source Software
# Purpose: Generate a personal open-source philosophy statement
# Inspired by: The Document Foundation's mission for LibreOffice

# Colors for better UI
GREEN='\033[0;32m'
BLUE='\033[0;34m'
YELLOW='\033[1;33m'
PURPLE='\033[0;35m'
NC='\033[0m' # No Color

# Clear screen for better presentation
clear

echo -e "${PURPLE}"
echo ""
echo ""
echo "    OPEN-SOURCE MANIFESTO GENERATOR"
echo "    Inspired by LibreOffice & The Document Foundation"
echo ""
echo ""
echo -e "${NC}"
echo ""
echo -e "${YELLOW}This manifesto generator asks you three questions about"
echo "your relationship with open-source software. Your answers"
echo "will be woven into a personal philosophy statement.${NC}"
echo ""
echo "Press Enter to continue..."
read

clear
```

```
# =====
# Collect user input
# =====

echo -e "${BLUE}===== ${NC}"
echo -e "${GREEN}Question 1 of 3${NC}"
echo -e "${BLUE}===== ${NC}"
echo ""
echo -e "${YELLOW}Name one open-source tool you use every day:${NC}"
echo -e "(Examples: LibreOffice, Firefox, Linux, Git, VLC, GIMP)"
echo ""
read -p "→ " TOOL
echo ""

echo -e "${BLUE}===== ${NC}"
echo -e "${GREEN}Question 2 of 3${NC}"
echo -e "${BLUE}===== ${NC}"
echo ""
echo -e "${YELLOW}In one word, what does 'freedom' mean to you in software?${NC}"
echo -e "(Examples: choice, transparency, control, privacy, community)"
echo ""
read -p "→ " FREEDOM
echo ""

echo -e "${BLUE}===== ${NC}"
echo -e "${GREEN}Question 3 of 3${NC}"
echo -e "${BLUE}===== ${NC}"
echo ""
echo -e "${YELLOW}Name one thing you would build and share freely:${NC}"
echo -e "(Examples: a document editor, a learning platform, a translation tool)"
echo ""
read -p "→ " BUILD
echo ""

# =====
# Get current date and user info
# =====
DATE=$(date '+d %B %Y')
USER_NAME=$(whoami)
HOSTNAME=$(hostname)
DISTRO=$(cat /etc/fedora-release 2>/dev/null || cat /etc/os-release | grep PRETTY_NAME | cut -d'"' -f2)

# =====
# Generate the manifesto
# =====
OUTPUT="manifesto_${USER_NAME}_${date +%Y%m%d}.txt"

cat > "$OUTPUT" << EOF
```

```
OPEN-SOURCE MANIFESTO
Generated: $DATE
By: $USER_NAME @ $HOSTNAME
System: $DISTRO
```

MY COMMITMENT

I, `$USER_NAME`, commit to:

- Using and supporting open-source software where possible
- Sharing my knowledge with others learning to code
- Contributing to at least one open-source project in the coming year
- Advocating for open standards and digital sovereignty

Signed,
\$USER_NAME
\$DATE

“Standing on the shoulders of giants, I will help build for those who come after me.”

EOF

```
# =====  
# Display the result  
# =====  
clear  
echo -e "${GREEN}☑ Manifesto generated successfully!${NC}"  
echo ""  
echo -e "${YELLOW}Saved to: $OUTPUT${NC}"  
echo ""  
echo -e "${PURPLE}=====
```

```
==${NC}"  
echo -e "${BLUE}YOUR OPEN-SOURCE MANIFESTO${NC}"  
echo -e "${PURPLE}=====
```

```
==${NC}"  
echo ""  
  
# Show the manifesto with line numbers for readability  
cat "$OUTPUT"  
  
echo ""  
echo -e "${PURPLE}=====
```

```
==${NC}"  
echo ""  
echo -e "${GREEN}Manifesto saved to $OUTPUT in the current directory.${NC}"  
echo ""  
  
# =====  
# Bonus: Create an alias suggestion  
# =====  
# The concept of aliases in shell scripting:  
# Aliases are shortcuts for commands. This script demonstrates  
# how aliases can be created to quickly generate future manifestos.  
  
echo -e "${YELLOW}💡 TIP: Add this alias to your ~/.bashrc for quick manifesto generation:  
${NC}"  
echo -e "    alias manifesto='$PWD/$0'"  
echo ""  
echo -e "Then run 'source ~/.bashrc' and type 'manifesto' anytime."  
echo ""  
  
echo "Press Enter to exit..."  
read
```

Screenshot

```

Manifest generated successfully!

Saved to: manifests_katvik_20200320.txt

=====

YOUR OPEN-SOURCE MANIFESTO

=====

      OPEN-SOURCE MANIFESTO
      Generated: 20 March 2020
      By: katvik @ katvik-wsl2t
      System: Linux Mint 21.3 Virginia

=====

MY COMMITMENT

I, katvik, commit to:
- Using and supporting open-source software where possible
- Sharing my knowledge with others learning to code
- Contributing to at least one open-source project in the coming year
- Advocating for open standards and digital sovereignty

MY PERSONAL PHILOSOPHY

I believe open-source software is about FREEDOM.
For me, freedom means control.

Every day, I use libreoffice to create and share my work.
This tool represents what open-source can achieve:
Software that respects its users and answers to no corporate agenda.

The Document Foundation taught us that when a project's future is threatened,
the community can fork, rebuild, and thrive.

MY PROMISE

One day, I will build a community document collaboration platform
and share it freely with the world. Not because it's easy,
but because others built tools for me, and I will build tools for others.

Signed,
katvik
20 March 2020

=====

"Standing on the shoulders of giants, I will help build for those who come
after me."

=====

Manifest saved to manifests_katvik_20200320.txt in the current directory.

💡 TIP: Add this alias to your ~/.bashrc for quick manifesto generation:
alias manifests="/home/katvik/manifesto_generator.sh"

Then run 'source ~/.bashrc' and type 'manifests' anytime.

Press enter to exit....

```

Explanation

This script demonstrates:

- **User input** with `read -p` for interactive prompts
- **String concatenation** using here-document (`<< EOF`) for multi-line output
- **File writing** with `>` and `>>` redirects
- **Command substitution** (`$(command)`) for dynamic values
- **Date formatting** with date `'+%d %B %Y'`
- **Variable usage** throughout for personalization
- **Color output** with ANSI escape codes
- **Alias concept** demonstrated via comment suggestion
- **Clear screen** with `clear` for better UX

The script asks three questions and generates a personalized open-source philosophy statement. This ties back to the course's core themes: why open-source matters, the four freedoms, and the responsibility to contribute.